Use work-energy to solve motion

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Learning target: Choose work-energy when force and displacement are known.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 2; 3 course days later.*
+24 J; force and displacement point the same way.

02. Explain a worked example (worked)
Net force along motion; energy grows from 0 to 12 J. Work-energy uses displacement without needing elapsed time.

03. Look and explain (model)
-18 J; final kinetic energy 0.

04. Build the solution (practice)
4.5 m.

05. Use the idea (reason)
Object likely stopped before assumed endpoint; braking model interval cannot continue with negative kinetic energy.

06. Choose a method (mixed) *Review: Unit 3, Day 7; 12 course days later.*
East positive: 6-4=2 kg·m/s east. Opposite directions require opposite signs.

07. Exit check (exit)
20 J.

08. STRETCH: Apply it another way (stretch)
Distances 6 and 3 m; larger opposing force stops it in less distance.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 2; 3 course days later.*
F=5 N; zero displacement gives 0 J.

02. Explain a worked example (worked)
Net force along motion; energy grows from 0 to 12 J. Work-energy uses displacement without needing elapsed time.

03. Represent and interpret (model)
-18 J; final kinetic energy 0.

04. Solve and justify (practice)
Initial 36 J; distance 12 m. Double speed gives 4 times energy/distance: 48 m.

05. Analyze the reasoning (reason)
Object likely stopped before assumed endpoint; braking model interval cannot continue with negative kinetic energy.

06. Choose a method (mixed) *Review: Unit 3, Day 7; 12 course days later.*
Deviation≈5%. It compares values relative to the initial value; uncertainty requires a measurement/error analysis.

07. Exit check (exit)
Net 40 J; v=√(80/5)=4 m/s.

08. STRETCH: Apply it another way (stretch)
Change in K=2-50=-48 J; force=-48/6=-8 N.